Feature taxonomy for addition at "="

How to read an operand plot

Each plot is one feature at the "=" token of the prompt "calc: a+b=". Vertical axis: a (first operand). Horizontal axis: b (second operand). Both run 0 to 99, so the plot covers all 10,000 prompts. Color: how strongly the feature fires on that prompt. White: it does not fire.

Directions in the plot

Top-left to bottom-right: a + b is constant (sum). Bottom-left to top-right: a - b is constant (the a = b diagonal is a - b = 0). Vertical lines: b is constant. Horizontal lines: a is constant.

Two labels per feature

Role (Anthropic's kinds): how the feature combines a and b, read from the shape. Family: what kind of condition it checks (last digit, magnitude, mod 100, comparison).

Step 1, the role: ask in order, stop at the first yes

1. More than half of the plot filled? -> mostly active. 2. Diagonal lines or bands, top-left to bottom-right? -> sum. 3. Only vertical bars, or only horizontal bars? -> add function. 4. L-shapes, corners, nested squares, or the a = b diagonal? -> comparison. 5. A regular grid, or a few separate blobs? -> lookup table. 6. None, several mixed, or a shape only in part of the plot? -> unclear.

Step 2, the family

last digit: the pattern repeats every 10. magnitude: one or a few smooth regions. mod 100: diagonal lines exactly 100 apart, or one thin diagonal line. comparison: about which operand is larger, or their difference. unclear: none of these.

How the code decides (for reference)

For each candidate variable v(a, b) the code computes R^2: the share of the feature's variation explained by the average activation within each value of v. It does this on the activation values and on the on/off pattern, and keeps the better one. A feature gets the category of its best variable only if R^2 >= 0.6 and it leads every other category by >= 0.15. A variable that contains a simpler one (e.g. (a+b) mod 100 contains (a+b) mod 10) must beat it by 0.05, and a joint lookup must beat the additive model (effect of a + effect of b) by 0.1. Otherwise: mixed or unclear.

When unsure

Write unclear and say why in the note column. A careful "unclear" is better than a guess.

add function / magnitude

Math

`v(a, b) depends on one operand only:  1[b in [lo, lo+L)],  floor(b/10)  (or the same for a); also g1(floor(a/10)) + g2(floor(b/10)) with no interaction between them.`

What it means

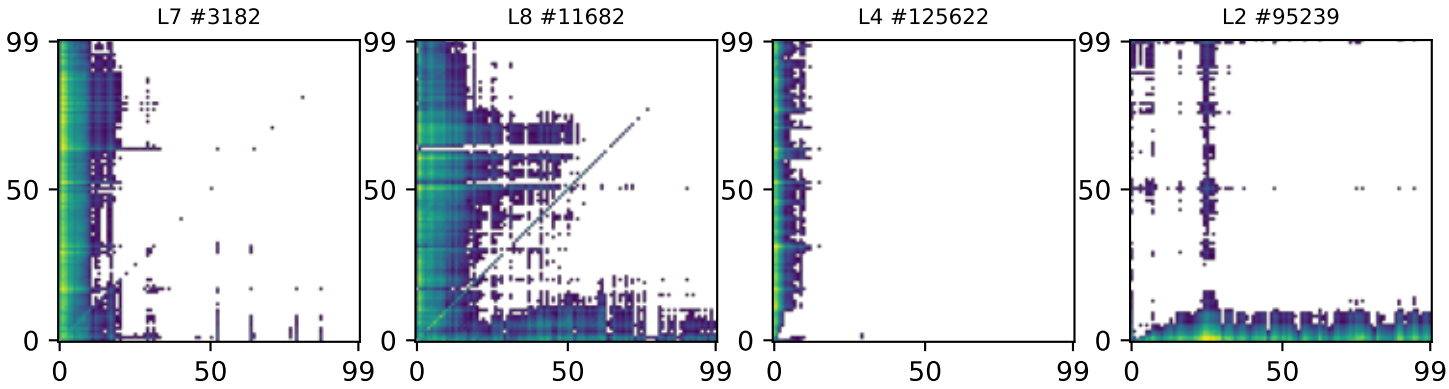
Reads the size of one operand. In Llama at "=" these are almost all "b is a single digit" (b < 10), so they track the operand's format more than its arithmetic size.

How it looks

A vertical bar (b fixed range, any a) or a horizontal bar (a fixed range, any b).

Confused with

Mostly active features with stripes (those fire almost everywhere; check how much is filled).



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

sum / last digit

Math

`v(a, b) = (a + b) mod 10`

What it means

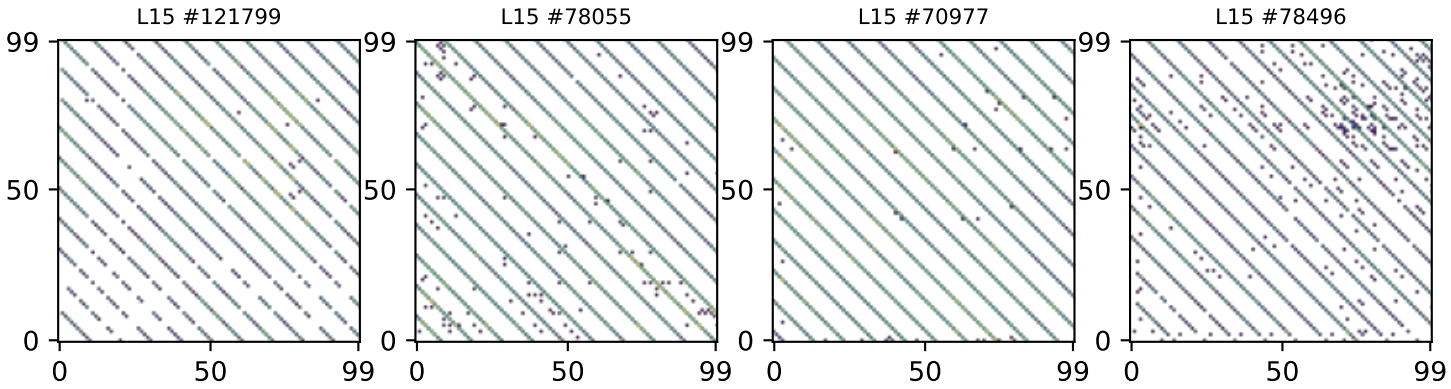
The units digit of the answer: 36 + 59 = 95 -> 5. This is the output side of the last-digit pathway.

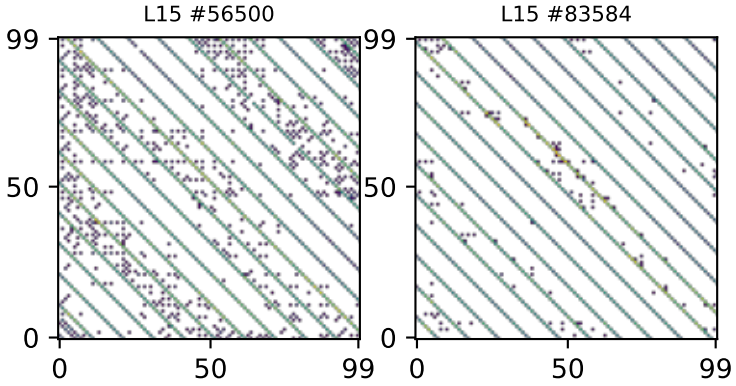
How it looks

Thin diagonal lines (top-left to bottom-right), one every 10 steps of a + b.

Confused with

mod 100 features: count the gap between lines. Every 10 -> last digit; 100 apart -> mod 100.





a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

sum / mod 100

Math

`v(a, b) = (a + b) mod 100`

What it means

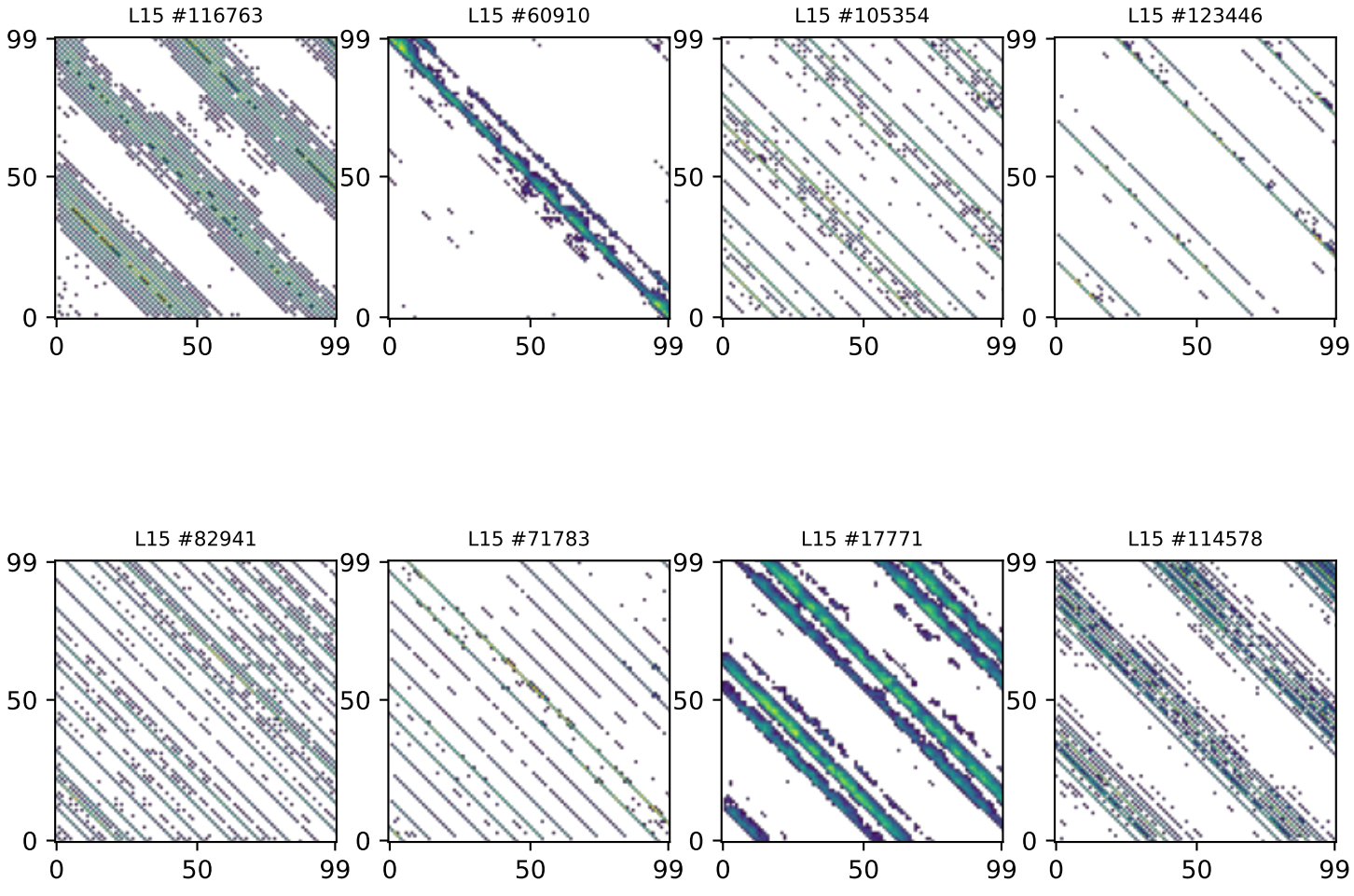
The last two digits of the answer: 95 and 195 are the same value (95). Fits the period-100 components found in the number embeddings.

How it looks

Diagonal lines or bands exactly 100 apart (e.g. a + b = 55 and 155), or one thin line (an exact sum below 100, e.g. a + b = 99).

Confused with

sum / magnitude: a single wide band is a range of sums, not mod 100.



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

sum / magnitude

Math

`v(a, b) = 1[a + b in [lo, lo+L)]  or  floor((a + b)/10)`

What it means

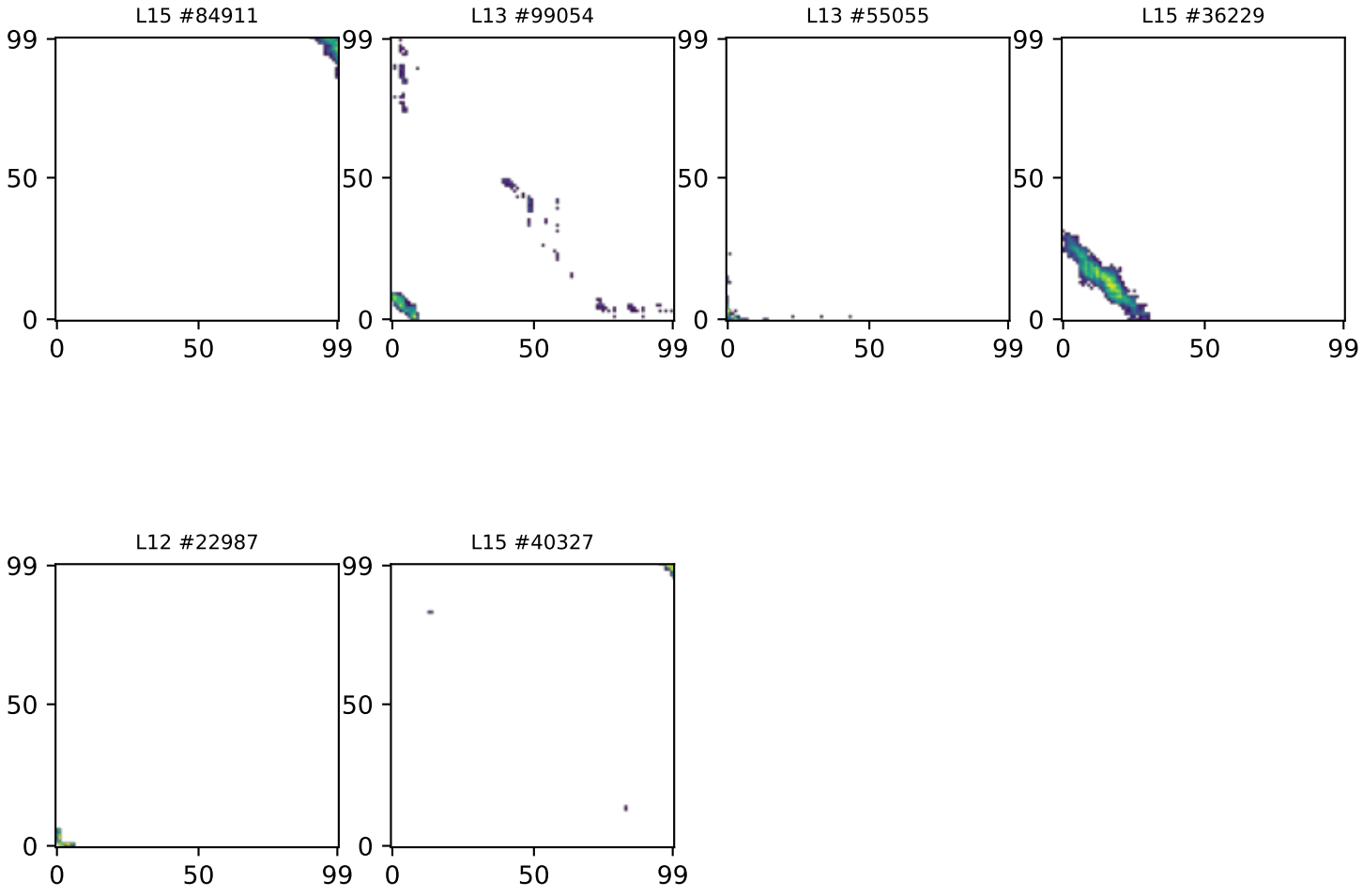
The rough size of the answer: "the sum is about 30". The low-precision side of addition.

How it looks

One wide diagonal band, or a corner (bottom-left: small sums; top-right: sums near 198).

Confused with

mod 100 (several lines 100 apart) and mostly active (a band that fills over half the plot).



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

lookup table / last digit

Math

`v(a, b) = (a mod 10, b mod 10) jointly: 100 combinations (e.g. "_6 + _9");`
`also carry = 1[(a mod 10) + (b mod 10) >= 10].`

What it means

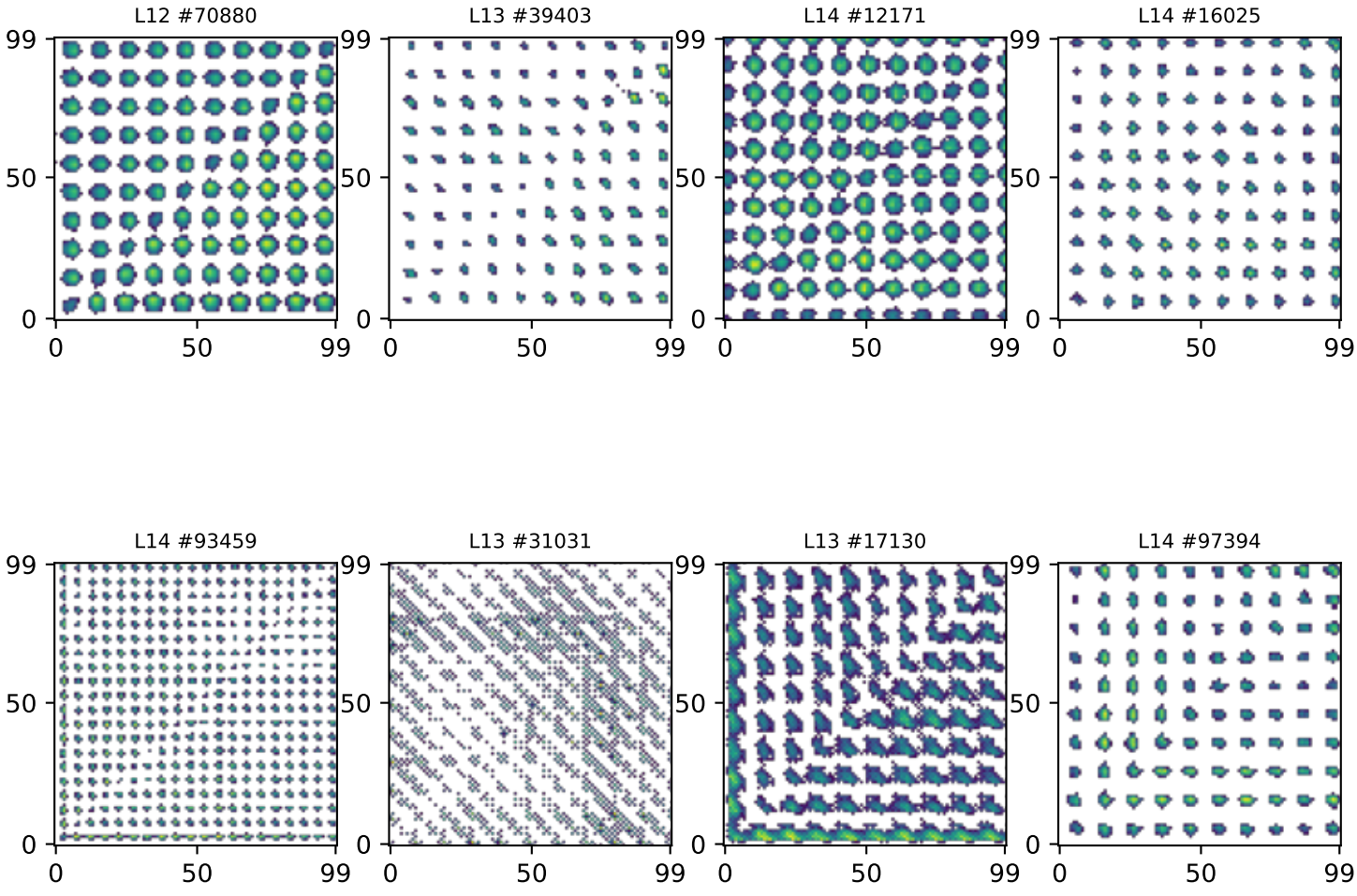
Combines the two units digits before the sum is formed, like a times table for units. Anthropic's "_6 + _9" features. Part of the last-digit pathway.

How it looks

A regular grid of dots or blobs repeating every 10 in both directions.

Confused with

Dense dot textures that fill the plot (mostly active), and nested squares (comparison / last digit).



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

lookup table / magnitude

Math

`v(a, b) = (floor(a/10), floor(b/10)) jointly, and it must clearly beat the additive model g1(floor(a/10)) + g2(floor(b/10)): a real interaction between a and b.`

What it means

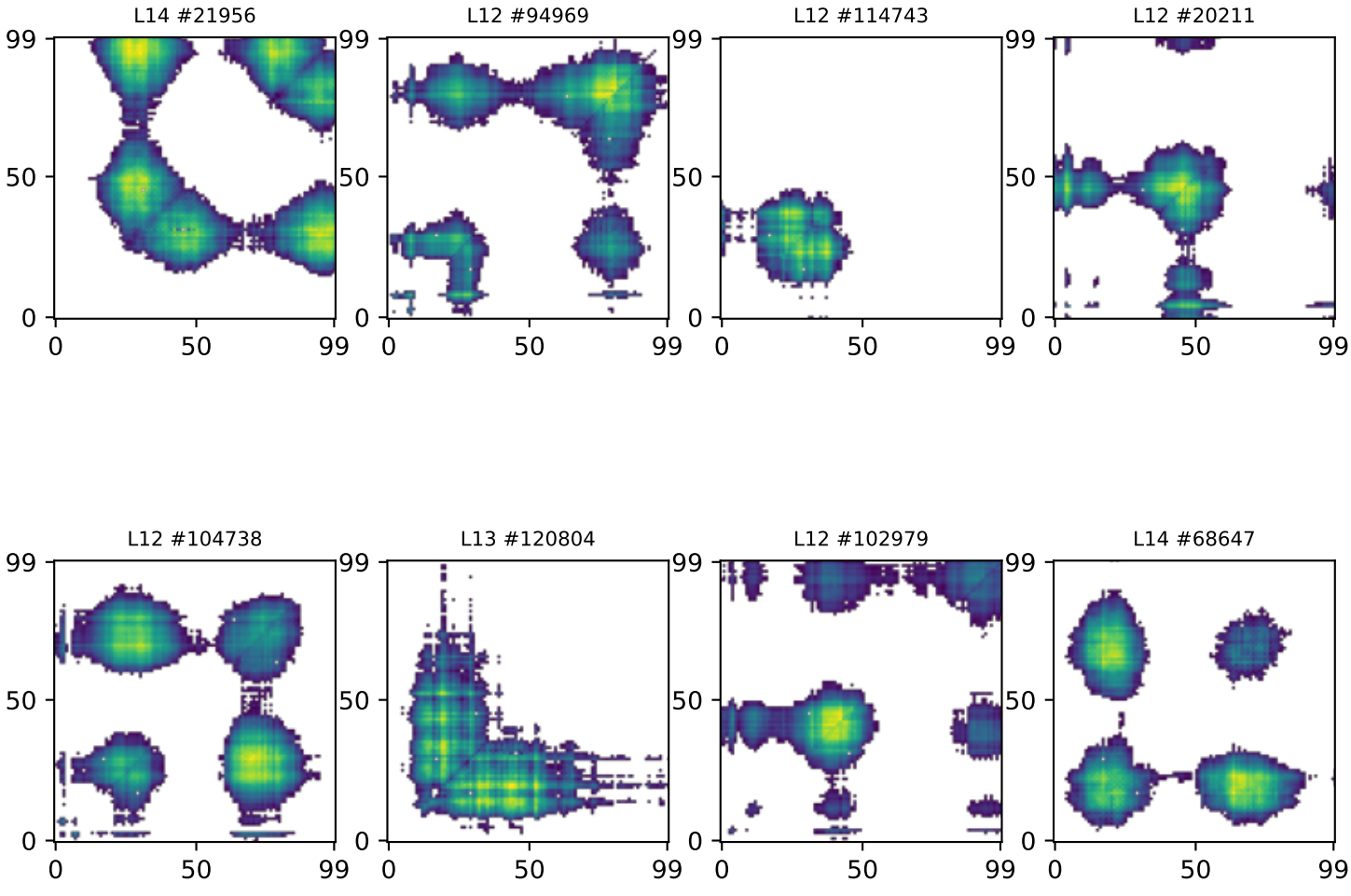
"a is about X AND b is about Y": a low-precision lookup (Anthropic's "~36 + ~60").

How it looks

One or a few separate blobs at specific (a, b) locations.

Confused with

L-shapes that bend at the a = b diagonal (comparison), wide diagonal bands (sum). This category had the most non-textbook candidates: be strict.



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

comparison / comparison

Math

`v(a, b) = min(a, b) or max(a, b) in a range or in decades;  a - b in a range;  1[a < b];  1[a = b]`

What it means

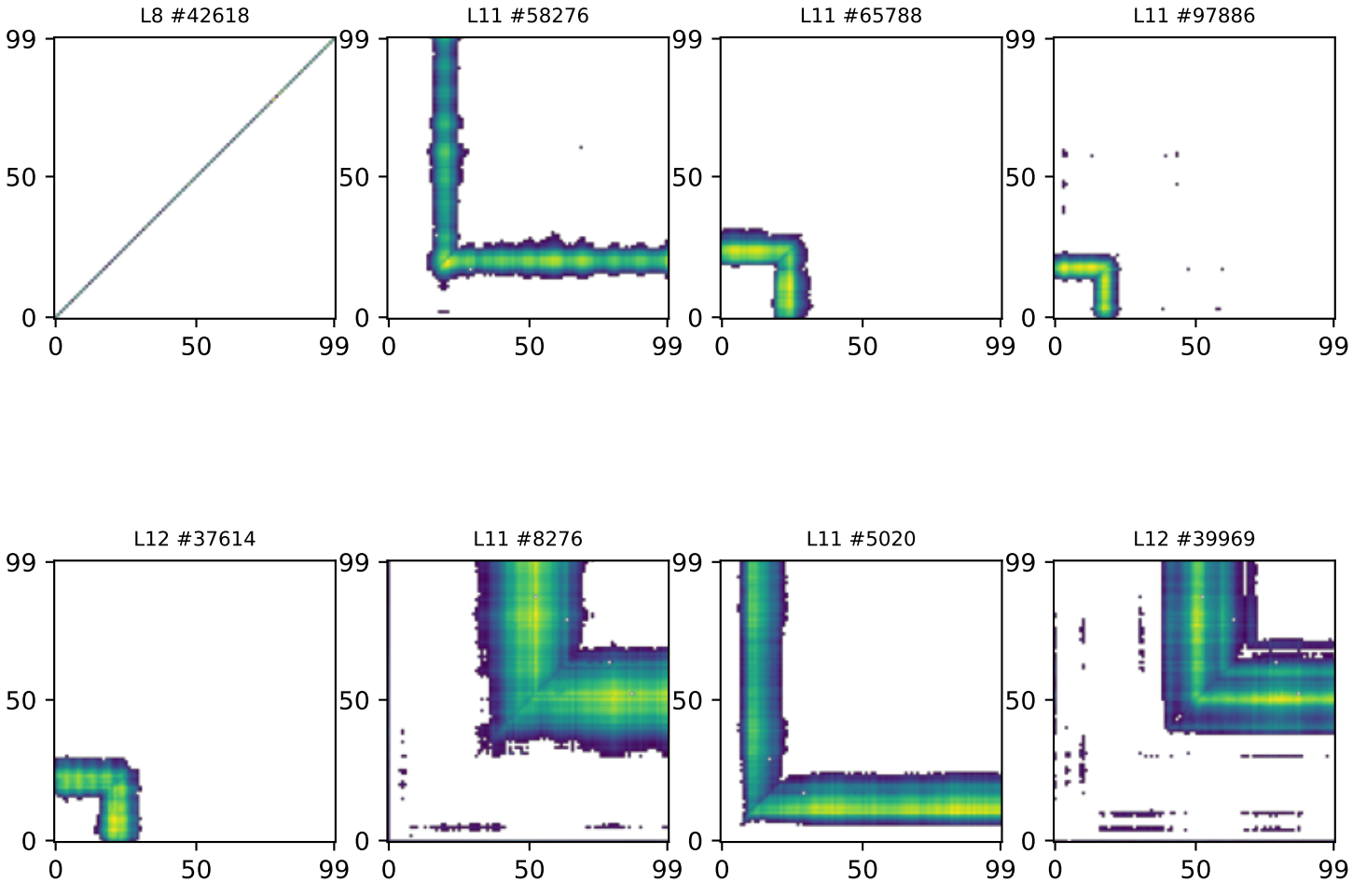
Which operand is larger, or how far apart they are: "the smaller operand is about 20".

How it looks

L-shapes or corners that bend exactly at the a = b diagonal; the a = b diagonal itself (bottom-left to top-right); a region above or below that diagonal.

Confused with

Blobs (lookup table). An L-shape whose corner lies on a = b is comparison.



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

comparison / last digit

Math

`v(a, b) = max(a, b) mod 10  or  min(a, b) mod 10`

What it means

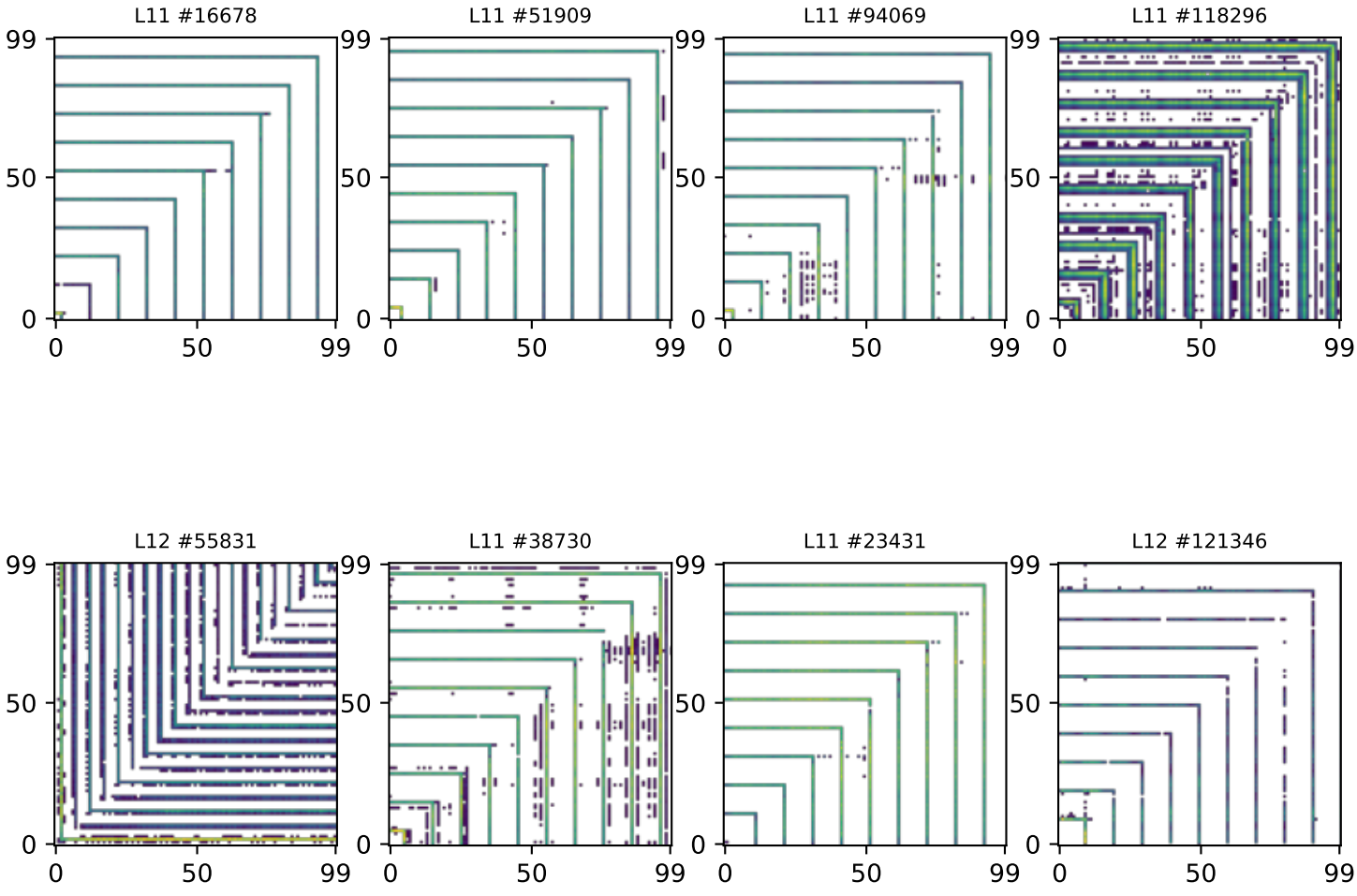
The last digit of the larger (or smaller) operand. Your pilot found "max(a, b) ends in d".

How it looks

Nested squares: lines that turn a corner at the a = b diagonal, one every 10.

Confused with

Lookup grids (dots in rows and columns, no corners).



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

mostly active

Math

`n_prompts / 9,992 > 0.5   (Anthropic: "active on the '=' token of most of our 10,000 prompts")`

What it means

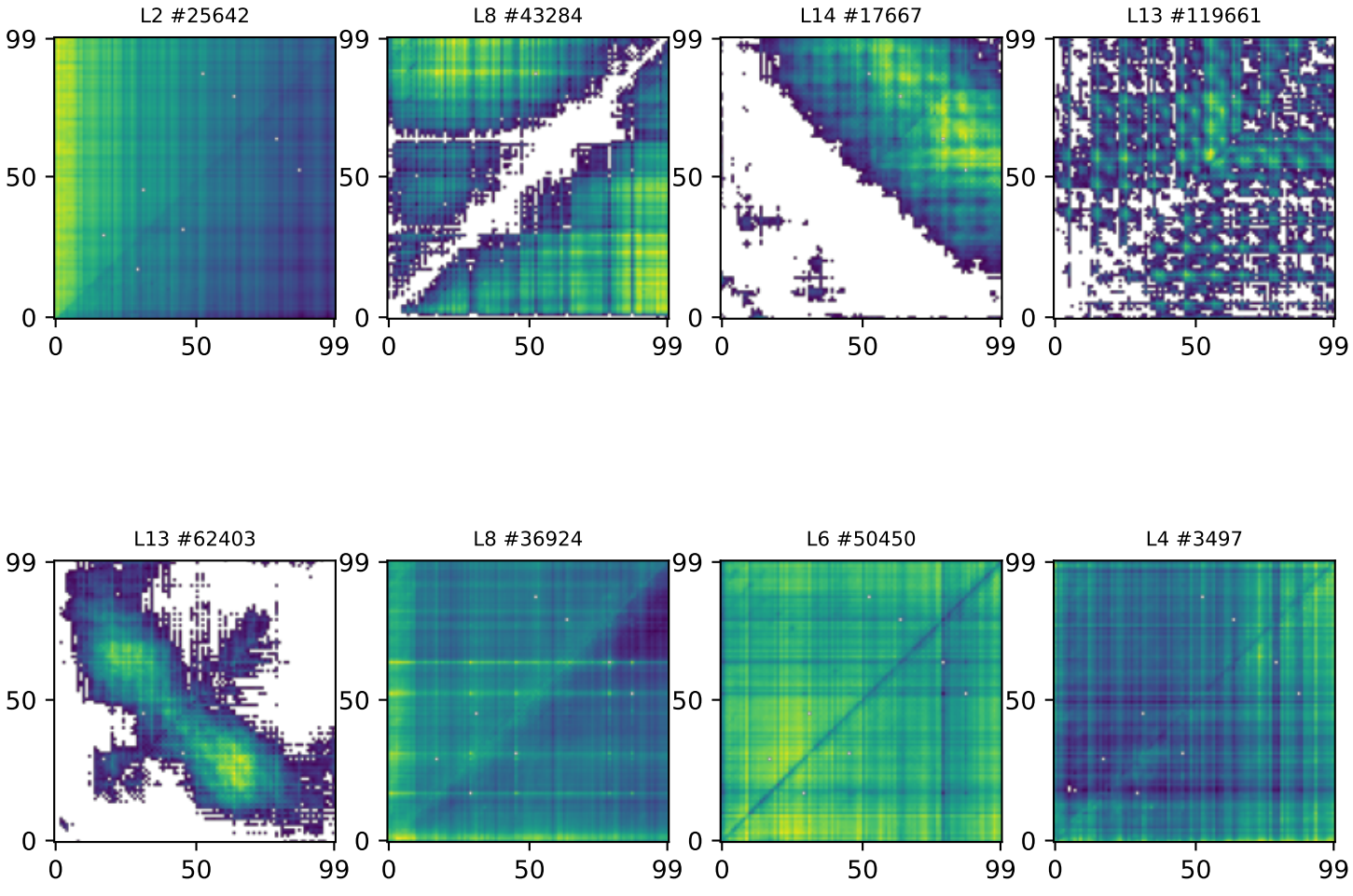
A feature that is on for most addition prompts. Its value can still vary a little with a or b, but its shape says nothing, so it gets this role whatever it looks like.

How it looks

More than half the plot filled, often with a texture or a faint a = b diagonal.

Confused with

A sum or comparison shape drawn inside a mostly filled plot is still mostly active.



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

unclear

Math

No variable reaches R^2 >= 0.6, or two categories score about the same ("mixed").

What it means

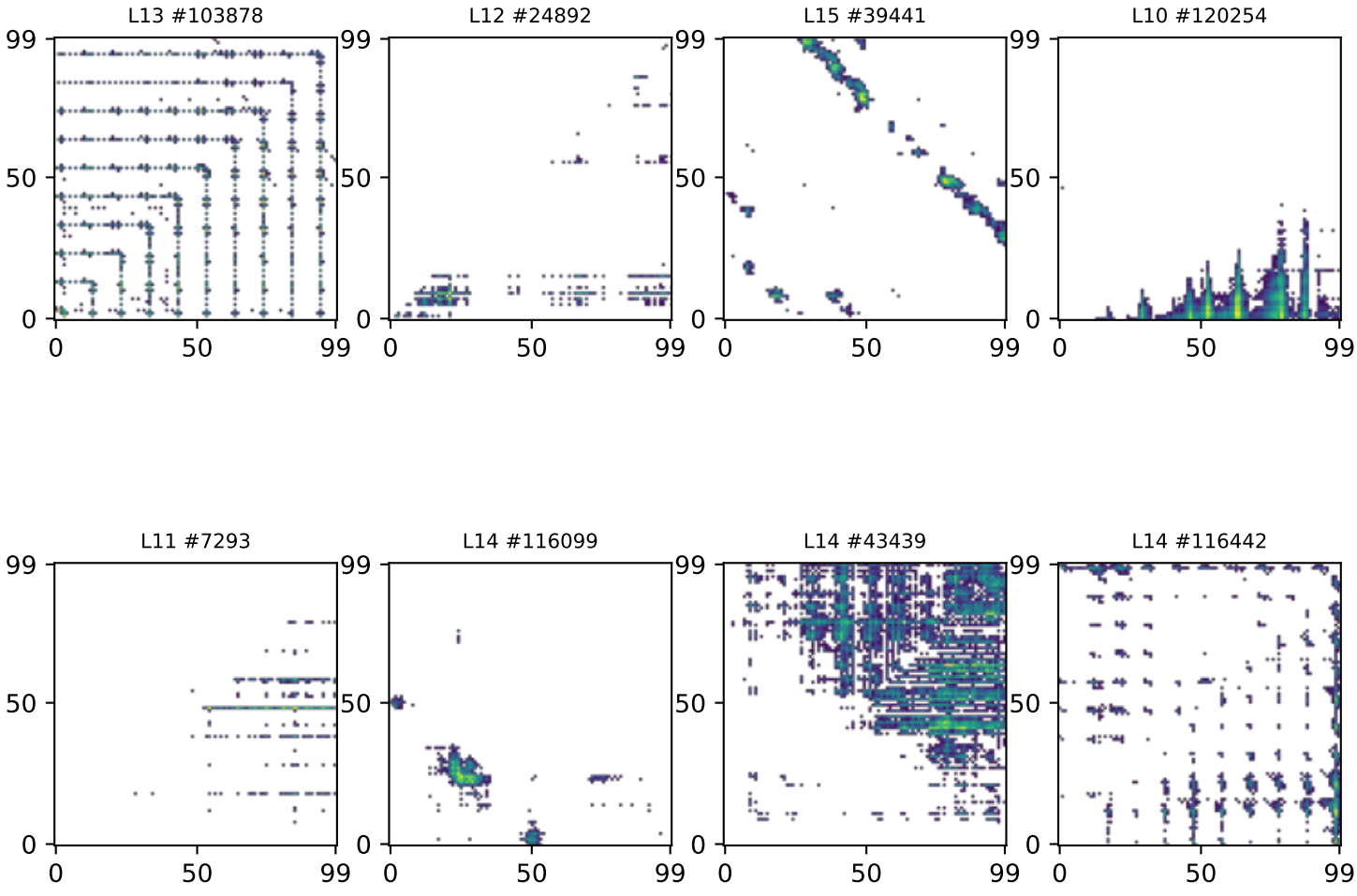
Either noise (scattered specks), two conditions at once (a lattice only in part of the grid), or a kind the variable library does not cover.

How it looks

Specks, several shapes mixed, or a clear shape that only covers part of the plot.

Confused with

Nothing: when in doubt, this is the right answer.



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.